

Query not attacker(messageV[]) is true.

Query not attacker(messageR[]) is true.

(* **Secrecy of session-key-protected communications:** application messages remain secret in both vehicle-to-RSU and RSU-to-vehicle directions.*)

Non-interference messageV is true.

Non-interference messageR is true.

(* **Strong message secrecy:** the attacker cannot distinguish executions carrying different application messages.*)

Query not attacker(IDi[]) is true.

(***Vehicle identity privacy:** the attacker cannot recover the vehicle's real identity from observable protocol messages.*)

Query inj-event(RSUAccept(g_3,rho_2,X_2,Y_2,k)) ==> inj-event(VehicleConfirm(g_3,X_2,Y_2,k)) is true.

Query inj-event(VehicleAccept(g_3,X_2,Y_2,k)) ==> inj-event(RSUResponse(g_3,rho_2,X_2,Y_2,k)) is true.

(* **Injective mutual authentication and key agreement:** each acceptance corresponds to a unique matching peer session agreeing on the service group, ephemeral public values, and session key.*)